

Native M5 as a realised variance source

Study III-B | Equivalence and coverage report
OANDA XAU_USD, already-viewed 2016-2023

Research question. Is native, unsmoothed five-minute midpoint data a defensible primary source for cross-session realised variance, given gaps in the available one-minute candles?

Finding

Yes, conditionally, for a new within-feed M5 RV methodology. All prespecified numerical checks pass globally and separately in every year. This does not establish independent-feed validity or equivalence of the full sample-selection procedure.

Audit result	Observed evidence
Shared complete-M1 blocks	535,892; all present in native M5
Bid/ask OHLC and price count	100% exact on shared complete blocks
Midpoint open/close discrepancy	At most \$0.0005 per ounce
Additional native M5 bars	30,542 with 1-4 received M1 candles
Native bars with no received M1	0
Permitted history	2016-2023 only; no fresh outcomes

Interpretation

Native M5 uses the same observed price-count mass and side extrema as the available M1 candles. It retains five-minute intervals that the five-of-five M1 rule discarded. More complete observation of the M5 process is defensible; describing these as recovered missing ticks would not be.

No trading strategy was run. No pre-2016 or 2024+ data was acquired. Study I, Study II and the existing Study III Design Pack remain separate and unchanged.

1. Source, clocks and measurement

The audit requested native XAU_USD M5 candles from OANDA practice HTTPS, with price=MBA and smooth=false, one UTC calendar day per request. Both bounds were explicit; the upper bound was exclusive. Original response bytes, SHA-256 hashes and request manifests are retained locally.

Source property	Value
Calendar requests	2,922; 2016-01-01 through 2023-12-31
Complete native M5 candles	566,434
Incomplete native candles	0
Empty calendar dates	439; includes non-trading dates
Native response bytes	155,880,589
Original M1 reference	2,789,384 complete paired bid/ask candles

Frozen session definitions

Asia is 09:00-15:00 Asia/Tokyo (00:00-06:00 UTC). London is 08:00-12:00 Europe/London: 08:00-12:00 UTC in winter and 07:00-11:00 UTC in summer. Windows are half-open, with historical timezone rules. UTC daily RV is a lagged control. Weekday coverage includes holidays; its denominator is not a claim that every weekday should trade normally.

M1 reconstruction and returns

A complete reconstructed M5 interval requires all five distinct M1 timestamps. Side OHLC is first open, maximum high, minimum low and last close; price-count volume is summed. Midpoint open and close are side-endpoint averages. RV is the unannualised sum of squared log close returns between consecutive received five-minute intervals, plus open-to-close for the first received block. No return crosses a missing M5 interval; no candle is filled or interpolated.

Native midpoint highs and lows are not reconstructed from side extrema. Averaging bid and ask highs or lows can combine observations from different instants. The original M1 archive contains BA, not true M OHLC, so a true M1-midpoint high/low equivalence test is unavailable.

Provider semantics: volume counts prices, not centrally traded ounces. A complete flag means the candle interval has ended; it does not certify five underlying M1 candles or tick-by-tick completeness. [1, 2]

2. Timestamp, OHLC and count equivalence

Year	Native M5	Complete M1 blocks	Partial-M1 native
2016	70,873	66,222	4,651
2017	70,570	64,772	5,798
2018	70,831	67,521	3,310
2019	70,707	59,855	10,852
2020	71,033	69,179	1,854
2021	70,855	68,637	2,218
2022	70,881	69,832	1,049
2023	70,684	69,874	810

Every complete block is contained

Timestamp containment is 100% in each year. There are no complete-M1 blocks absent from native M5, and no native M5 blocks with zero M1 observations. Shared timestamps are five-minute UTC starts. Source order, uniqueness and request-interval membership are checked.

Numerical agreement

All eight bid/ask OHLC fields agree exactly on all 535,892 complete blocks. Summed M1 price counts equal native M5 counts on every shared complete block. Native midpoint open and close differ from reconstructed side averages by at most \$0.0005, consistent with endpoint rounding; all are within the frozen \$0.001 tolerance.

Across 514,540 shared consecutive close-return edges, return correlation is 0.9999998808. The 95th percentile absolute log-return discrepancy is 5.560×10^{-7} ; the maximum is 9.407×10^{-7} . These are log-return units, not percentages.

True midpoint extrema

All native midpoint extrema lie within the corresponding bid/ask extrema bounds at the \$0.001 tolerance. However, averaging side highs/lows differs from true midpoint extrema: median absolute difference is \$0.0015; maxima are \$3.5915 for highs and \$3.2530 for lows. These diagnostics reinforce why side-extrema averages must not be labelled true midpoint highs/lows.

3. What exists during missing M1 intervals?

M1 candles in M5	Native bars	Price-count equality
1	619	100%
2	2,347	100%
3	5,693	100%
4	21,883	100%
5	535,892	100%

The 30,542 partial-M1 bars contain 42,786 missing constituent-minute slots. In every such bar, native bid/ask OHLC equals aggregation of the received M1 candles, and native price-count volume equals their sum. Thus native M5 adds usable aggregation intervals, but no additional recorded price-count mass relative to the original M1 archive.

Endpoint timing is part of the limitation

All: among 30,542 partial bars, 7,273 lack the opening M1 minute and 8,928 lack the closing M1 minute. The largest constituent-minute displacement is 4 minutes.

Asia: among 12,465 partial bars, 3,011 lack the opening M1 minute and 3,481 lack the closing M1 minute. The largest constituent-minute displacement is 4 minutes.

London: among 2,935 partial bars, 575 lack the opening M1 minute and 715 lack the closing M1 minute. The largest constituent-minute displacement is 4 minutes.

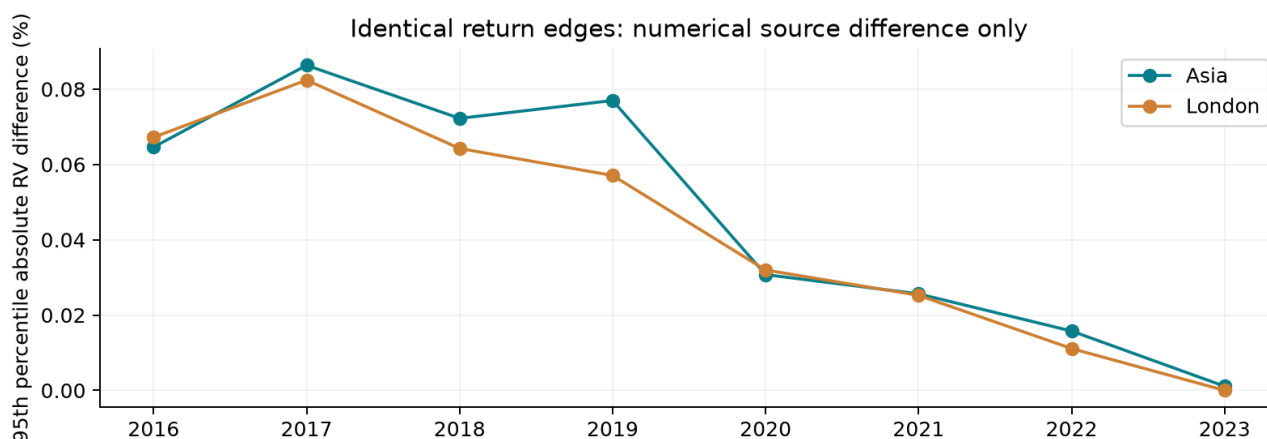
These are offsets of received M1 buckets, not measured tick ages. `smooth=false` uses the first available price in the interval for the open; the close reflects the last available price. A complete native bar can therefore be sparse within its five-minute bucket. Tick freshness and the economic reason for absent M1 records remain unverified. [1, 2]

What this resolves, and what it cannot resolve

The prior impression that M5 might recover wholly missing M1 intervals is too broad: there are no native bars in five-minute intervals with zero received M1 candles. The evidence is consistent with common sparse source observations and different completeness requirements. It does not prove that no quotes occurred in the market, identify a provider outage, or establish that missingness is random.

Fully empty dates and longer blank stretches are not repaired by switching candle resolution. The audit does not infer missing prices or treat volume equality as an independent feed check.

4. RV equivalence versus added observation



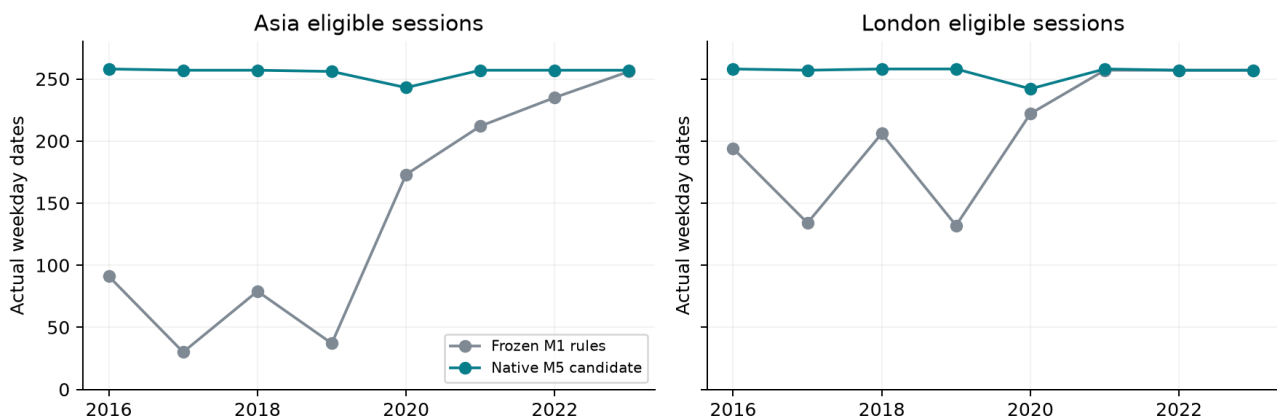
Comparison, all paired dates	Asia median / p95	London median / p95
Identical return edges relative difference	0.0102% / 0.0603%	0.0084% / 0.0532%
All native vs complete-M1 RV relative difference	2.1879% / 62.4717%	0.0164% / 27.8839%
All native vs complete-M1, old-valid dates	0.0099% / 4.6221%	0.0086% / 6.4203%

The identical-edge comparison isolates rounding and source numerics. The all-native comparison also changes the observed blocks and restores adjacent return edges lost when complete-M1 blocks were discarded. The latter is a different measurement/sample procedure, not a failure of the shared-bar equivalence result.

For all paired dates, signed mean RV changes are +15.815% in Asia, +6.018% in London and +3.563% for the UTC day. Relative changes can be large when reconstructed RV is very small; the maximum Asia change is about +1,739%. These tails are retained and reported, not winsorised.

RV reconstructed from the original M1 responses exactly reproduces frozen Study II RV on all its valid dates: 1,113 Asia, 1,659 London and 1,429 daily observations. This checks the measurement implementation without changing the old study.

5. Coverage and the candidate quality rule



The separately specified native candidate requires at least 95% of expected M5 session candles, both boundary blocks, maximum received-timestamp gap of 10 minutes, no quote flags and positive RV. Daily controls require 85% M5 and maximum gap of 90 minutes. Quote flags include invalid OHLC, nonpositive or greater-than-\$5 endpoint spreads, or an adjacent midpoint open/prior-close log jump above 2%. All 1,089 flagged native bars are recorded; no result-based deletion is used.

Session	Frozen eligible	Native candidate	Added / lost
Asia	1,113	2,042	929 / 0
London	1,659	2,045	386 / 0
UTC daily	1,429	1,944	515 / 0

These are alternative eligibility definitions, not corrected Study II counts. In particular, native M5 eligibility does not certify 98% M1 completeness. For 2020, widened-spread and other quality exclusions remain visible. Coverage gain is substantial but not a guarantee of unbiased RV.

No New York entry rule, forecast acceptance test on fresh outcomes, or trading translation is part of this report.

6. Decision and research consequences

Decision: native unsmoothed OANDA M5 midpoint is a defensible primary observable RV source for a separately registered within-OANDA study, subject to the stated sparse-bar and quote-quality controls. It is not independent-provider confirmation and should not replace frozen Study II outcomes retrospectively.

Year	Forecast-eligible	Eligible with London label
2016	237	237
2017	256	256
2018	255	255
2019	255	255
2020	217	213
2021	256	256
2022	256	256
2023	256	256

These new-method feature counts use current eligible Asia, prior valid London/day at most four calendar days old, and 5/20 prior valid London observations. They are coverage diagnostics only. The first 400-row training subtraction has not been used to reopen or rerun Study II validation. The full-procedure simulation uses 1,904 pilot rows with all current calibration primitives, including daily RV, available.

Before any fresh-period outcome study

Register the native-M5 estimand, sparse-candle policy, quality thresholds, calendar coverage gate and independent-feed role in a new design. Audit source availability before evaluating outcomes. The present equivalence report is the required sequencing step; it does not itself authorise or execute pre-2016 acquisition.

The companion power report rebuilds planning with recursively generated variances, 400-row model fitting, frozen coefficients and actual forecast losses. Its synthetic results cannot validate a trading strategy or establish a new empirical market effect.

7. Reproducibility and references

This phase is isolated in study-iii-b. The data-method plan was committed before native acquisition; the full simulation algorithm was frozen separately before power execution. Prior-study source and artifact hashes were captured before analysis. Final delivery includes a verification ledger and a research bundle of code, frozen plans, tables and reports. Raw provider responses and credentials are not included in the portable bundle.

Audit material	Purpose
M5_SOURCE_INVENTORY.csv	All daily request hashes, counts and bounds
TIMESTAMP / FIELD / RETURN_AGREEMENT.csv	Global and annual numerical comparisons
NUMERICAL_GATES_BY_YEAR.csv	Unchanged tolerances, all years and pooled
PARTIAL_BLOCK_OHLC_AGREEMENT.csv	Received-M1 side aggregation, counts 1-5
CONSTITUENT_ENDPOINT_TIMING.csv	Bucket endpoint offsets; not tick ages
SESSION_EQUIVALENCE_LEDGER.csv	Date/session RV and eligibility comparison
FROZEN_RV_REPRODUCTION.csv	Exact old-valid RV reproduction
PRIOR_ARTIFACT_VERIFICATION.json	Frozen-study preservation check

Sources

[1] OANDA. Instrument definitions: Candlestick, CandlestickData and granularity. developer.oanda.com/rest-live-v20/instrument-df/

[2] OANDA. Official v20 Instrument API schema: price components, smooth, from/to and candle delivery. [Official v20 instrument schema](#).

[3] Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. Journal of Econometrics 160, 246-256. [Author-hosted paper](#). Proxy-robust loss results do not make arbitrary sparse quote sampling unbiased.

[4] Local frozen research: Study II OANDA report and source archive; Study III Design Pack; this phase's DATA_METHOD.md and FULL_PROCEDURE_SIMULATION.md. Their original decisions and acceptance requirements are unchanged.

Scope attestation

Only actual OANDA 2016-2023 data are described as observed market evidence. Synthetic power paths are explicitly labelled synthetic. No pre-2016 or 2024+ market data, trade outcomes, strategy optimisation, EA implementation or C++ work are included.